
Romanization Flattens the Field: Script Variation and Metric Choice in Sinhala Evaluation

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Abstract

1 Benchmarks for Global South languages are written in the formal script, while the
2 people who would use a model mostly type an ad hoc Romanized register in Latin
3 letters. We study that mismatch in Sinhala, an Indo-Aryan language with more
4 than 17 million speakers. A recent benchmark reported that language models are
5 about 312 times worse on Romanized than on Sinhala-script text, and that model
6 scale predicts nothing. Re-measuring 31 checkpoints with tokenizer-independent
7 bits per byte, we reproduce the published perplexities and split the reported ratio
8 exactly in two: a factor of 21 is identical content being cut into a different number
9 of tokens, and a factor of 24 is a real rise in per-byte loss. Under the corrected
10 metric, scale does predict Sinhala-script ability. In the first downstream evaluation
11 of Sinhala script variation, 9,479 items and eleven instruction-tuned checkpoints
12 in both scripts for 208,538 generations, romanization does not lower a field of
13 models uniformly, it flattens it, and a 22.4 point spread in Sinhala-script accuracy
14 becomes 6.7 points. On offensive-language detection, accuracy barely moves while
15 Matthews correlation loses more than half its value. We close with the evaluation
16 practices this implies for other languages in the same position.

17 1 Introduction

18 Sinhala is an Indo-Aryan language written in a script of its own and spoken by more than 17 million
19 people, almost all of them in Sri Lanka [de Silva, 2019, Ranasinghe et al., 2025]. Like several South
20 Asian languages it has an everyday orthographic split: formal writing uses the Sinhala script, while
21 comments, chat and social media are dominated by Romanized Sinhala, known locally as Singlish
22 and typed with Latin letters. It has no standard spelling, so `[sin:kohomada]` (“how are you”) is typed
23 *kohomada*, *kohomd* or *kohomade* depending on the writer.

24 Every published Sinhala evaluation set we know of is in the Sinhala script. That is an easily missed
25 coverage gap, because the benchmark is in the right language and the wrong register. Romanized
26 registers persist for durable infrastructural reasons rather than as a fashion [Sengupta et al., 2024],
27 and in a deployed maternal-health service Khullar et al. [2025] found 83% of Hindi and 73% of
28 Telugu user messages Romanized, with models handling them measurably worse. Anyone serving
29 Sinhala users is serving mostly Romanized input while validating on Sinhala-script data.

30 Rajapakse and Weerasinghe [2026] measured that gap first, benchmarking 24 open-weight models
31 with perplexity and reporting a median 312-fold degradation from Sinhala-script to Romanized text
32 and no correlation between parameter count and script handling. Both numbers depend on the metric
33 rather than on the models, because perplexity is normalised by token count and token count for a
34 fixed string is entirely a property of the tokenizer [Gao et al., 2020, Magnusson et al., 2024]. Sinhala
35 is exactly where that matters, since tokenizers segment it very unevenly [Petrov et al., 2023, Ahia
36 et al., 2023].

Recomputing tokenizer-independent bits per byte for 31 checkpoints splits the reported ratio exactly in two, a factor of 21 from token-count normalisation and a factor of 24 from a real rise in per-byte loss, and the claim that scale does not help does not survive (§3). In the first downstream evaluation of Sinhala script variation, 9,479 items and 208,538 generations, romanization compresses the range of model quality far more than it shifts the mean, turning a 22.4 point spread in accuracy into 6.7 points (§4). And on offensive-language detection, accuracy falls a few points while Matthews correlation loses more than half its value, so the metric decides whether the damage is visible at all. None of this needed new models or new annotation, only a different denominator and a competence screen. §5 states those as practices.

2 Setup

Data and conditions. Every item is evaluated twice, in the Sinhala script as published and in a Romanized form with identical content, so every comparison is paired and differs only in orthography. The intrinsic corpus is the 500 parallel sentence pairs of Rajapakse and Weerasinghe [2026], whose Romanized side native speakers typed, so the intrinsic results do not depend on our transliteration. Downstream we use every item of three sets with no sampling: 6,879 multiple-choice items from SinhalaMMLU [Pramodya et al., 2025], where a mix of four and five option items puts guessing at 23.4%, the complete 2,500-item test split of the SOLD offensive-language task [Ranasinghe et al., 2025], 40.6% offensive so the majority class scores 59.4%, and the 100 Sinhala items of Global PIQA [Chang et al., 2025, de Silva and Ranathunga, 2026]. That is 9,479 items over eleven instruction-tuned checkpoints from 1.1B to 14.7B, or 208,538 zero-shot greedy generations in 80 GPU hours, with one prompt template for every dataset so task and prompt are not confounded. The intrinsic sweep covers 31 checkpoints from 350M to 14.7B, 24 of them prior work’s pool (Appendix B).

The Romanized side. Almost every Sinhala transliteration tool runs Latin to Sinhala [Sumanathilaka et al., 2025] and we need the reverse, so we wrote a deterministic grapheme-level transliterator using the digraph conventions speakers actually type, turning `[sin:wesak uthsawayeedii bauddhayin]` into *wesak uthsawayeedii bauddhayin*. Against 755,000 human-romanized reference items it has the lowest character error rate of four methods, and 88.6% of its spellings are ones a human supplied for that word, against 84.2% for the speakers who typed the same sentences (Appendix H). It is released with the evaluation harness and all analysis code at <https://anonymous.4open.science/r/XXXXXXX>.

Statistics and the competence screen. Paired comparisons use McNemar’s test [McNemar, 1947] with Holm correction over checkpoints within a task [Holm, 1979], and intervals come from 10,000 paired bootstrap resamples. A model already at chance in the Sinhala-script condition cannot lose anything by being given Romanized input, so averaging it in would understate the effect. We mark a checkpoint *competent* on a task when its Sinhala-script score beats the guessing baseline at $p < 0.01$ under a Poisson-binomial null and it does not put more than 90% of its answers on one option. Six of eleven pass on SinhalaMMLU, six have a real signal on SOLD, and none pass on Global PIQA.

3 The Metric Was Measuring the Tokenizer

Bits per byte divides total negative log-likelihood by a unit of the text itself rather than by its tokenization, $\text{BPB} = \sum_s \text{NLL}(s) / (\ln 2 \sum_s |s|_{\text{bytes}})$, and likewise per whitespace word, so the denominator no longer depends on the vocabulary. Our recomputed perplexities reproduce the published values to a median of 0.05%, so what follows is the metric and not a different run (Appendix C).

The two metrics disagree. Across the 31 checkpoints, Sinhala-script perplexity and bits per byte are statistically unrelated ($\rho = 0.08$, $p = 0.68$), and Gemma-2-9B ranks 26th of 31 on the first and 2nd on the second. Perplexity tracks tokenizer fertility instead ($\rho = -0.87$, $p < 10^{-9}$), which runs from 4.3 tokens per Sinhala word for the Qwen3.5 and Gemma tokenizers to 14.2 for models that fall back to bytes, and spreading the same loss over three times as many tokens lowers perplexity without the model knowing more.

That makes the reported ratio decomposable rather than arguable. Perplexity is $\exp(\text{NLL}/T)$ for T scored tokens and $\text{NLL} = \ln 2 \cdot A \cdot B$ for A bits per byte over B bytes, so $\log_2 \text{PPL} = Ax$ with

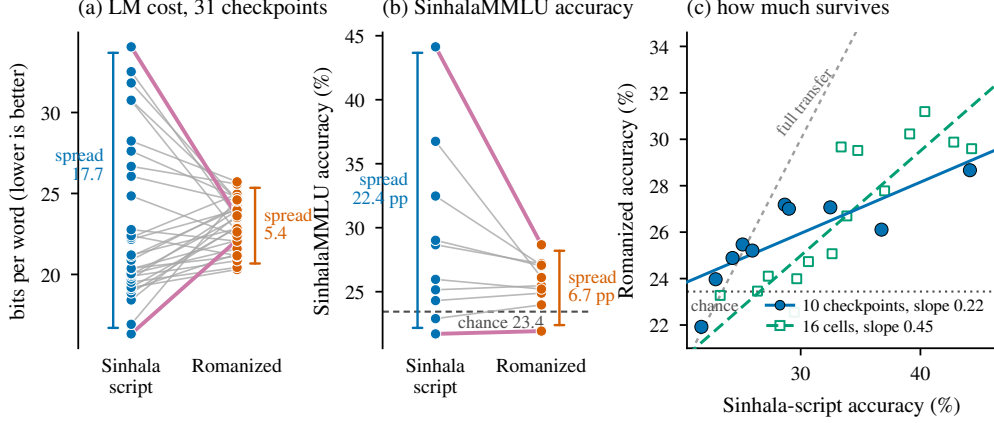


Figure 1: (a) Language modelling cost. Word counts are identical within a pair, so the columns are comparable, and the highlighted lines are checkpoints cheaper on Romanized text. (b) SinhalaMMLU accuracy, where a 22.4 point spread becomes 6.7. (c) The same numbers as levels, with the 16 domain by difficulty cells. Both slopes fall well below the full-transfer slope of 1. LaMini-GPT-1.5B is omitted from (b) and (c) for unparseable output on 95.9% of Sinhala-script items.

88 $x = B/T$ the bytes each token carries, and the change across script conditions is exactly

$$\Delta \log_2 \text{PPL} = \underbrace{\bar{A}(x_r - x_u)}_{\text{token-count normaliser}} + \underbrace{\bar{x}(A_r - A_u)}_{\text{per-byte loss}}, \quad (1)$$

89 with $\bar{A}$ and $\bar{x}$ the two-condition means. The first term is identical content being cut into a different
90 number of tokens, the second the model genuinely assigning less probability per byte. This is an
91 identity rather than a model, and it holds to 5×10^{-15} bits in our data.

92 On prior work’s 24 checkpoints the median gap is 8.29 bits, which is their 312-fold ratio, and the
93 split assigns 4.37 bits to the normaliser and 4.56 to per-byte loss, a factor of 21 and a factor of
94 24 whose product is 312. Almost exactly half of the headline is the unit of account, and the same
95 measurement stated tokenizer-independently is 2.14 bits per byte, a factor of 4.4. The scale claim
96 reverses too. Parameter count is uncorrelated with Sinhala-script perplexity ($\rho = 0.15$, $p = 0.42$) but
97 correlated with Sinhala-script bits per byte ($\rho = -0.55$, $p = 0.001$), including on prior work’s own
98 24 checkpoints ($\rho = -0.42$, $p = 0.04$). What scale does not buy is Romanized ability ($\rho = -0.28$,
99 $p = 0.13$). Family, tokenizer and corpus vary together here, so that is an association and not a scaling
100 result.

101 4 Romanization Flattens the Field

102 Figure 1(a) shows the effect that organises the rest. Sinhala-script cost spans 16.3 to 34.1 bits per
103 word across the 31 checkpoints while Romanized cost collapses into a 20.3 to 25.7 band, and a full
104 bit per word of Sinhala-script improvement buys only 0.14 bits on Romanized text (95% CI 0.05 to
105 0.25). Twelve of the 31 are *cheaper* on Romanized text, and which ones is predictable, because the
106 better a checkpoint is on the Sinhala script the more it loses ($\rho = -0.88$, $p < 10^{-9}$). No checkpoint
107 beats a held-out character bigram on Romanized Sinhala, 3.06 bits per byte against 3.29 for the best
108 of the 31 and 3.81 for the median (Appendix D).

109 **The same shape appears downstream.** Four of the six competent checkpoints lose significantly on
110 SinhalaMMLU after correction (Appendix Table 7), Qwen3.5-9B falling from 44.1% to 28.7% (95%
111 CI 14.0 to 16.9), and the two that lose least had 2.5 and 5.2 points of headroom. Pooled, the gap is
112 6.0 points (95% CI 5.4 to 6.5, $n = 41,274$). Figure 1(b) shows the levels, a 22.4 point Sinhala-script
113 spread becoming 6.7, with no checkpoint above 28.7% against a 23.4% chance baseline, so the best
114 one keeps 5.2 of its 20.7 points of headroom. Regressing Romanized on Sinhala-script accuracy gives
115 a slope of 0.22 (95% bootstrap CI 0.11 to 0.54, leave-one-out 0.17 to 0.26), reliably below the 1.0 of
116 full transfer ($p = 8.6 \times 10^{-7}$) and above the 0.0 of total flattening. A refit over the benchmark’s 16
117 domain by difficulty cells, where the units are items, gives 0.45 (95% CI 0.36 to 0.63).

118 **On safety, accuracy hides it.** Accuracy on SOLD falls by 1.6 to 11.3 points against a 59.4%
119 majority baseline, which looks survivable, while the median checkpoint keeps only 45% of its
120 Matthews correlation. Five of the six lose significantly. The mechanism is a retreat toward triviality,
121 and Qwen2-7B-Instruct more than doubles its rate of predicting the offensive class, 10.5% to 23.4%,
122 without becoming more accurate. A deployment tracking accuracy would see a five point wobble
123 while over half the usable signal disappeared.

124 The effect is neither a tokenization artifact nor output degeneration. All eleven checkpoints receive
125 *fewer* tokens in the Romanized condition on all three tasks, a median 2.09 per word against 9.5 for
126 the Sinhala script, and the invalid rate never exceeds 0.03% in either condition (Appendix E). The
127 models keep answering across the option set and are simply wrong.

128 5 What This Means for Evaluating Global South Languages

129 Sinhala is one language, but the instruments that failed here are the ones in general use and the
130 conditions that made them fail are common.

131 **Report a tokenizer-independent number.** A benchmark comparing perplexity across model
132 families, on a script those families tokenize unevenly, reports tokenizer behaviour as much as model
133 behaviour. Non-Latin and low-resource scripts are precisely where fertility varies most [Petrov et al.,
134 2023], so for our languages that is the default case and not a corner case. Bits per byte costs nothing
135 extra on the same forward passes, and here it moved a headline by a factor of 21.

136 **A benchmark can be in the right language and the wrong script.** Coverage is usually counted
137 in languages, but for any language with a dominant Romanized register, and Hindi, Telugu, Urdu,
138 Nepali and Arabic dialects all qualify to some degree, a language count overstates coverage of what
139 users type. Building the second condition here needed a transliterator in the direction almost no
140 tooling supports, which is itself a reason these registers stay unmeasured.

141 **Screen for competence and state the guessing baseline.** Averaged over our eleven checkpoints
142 the script effect looks modest, because five of them are at or near chance natively and contribute
143 a zero gap by construction. For the same reason the hardest strata reveal the smallest effects, 7.9
144 points on Easy items against 4.3 on Hard. Low-resource evaluation is full of near-chance results, so
145 unscreened group means understate script and dialect effects rather than overstate them.

146 **Choose a metric that survives class imbalance.** The safety task here is 40.6% positive, so accuracy
147 is anchored by the majority class. That is the normal shape of moderation and triage data, and it is
148 where a metric can look fine while the system has stopped working.

149 **The ranking you validate on is not the ranking you deploy.** Because flattening is proportional
150 to ability, the checkpoints that look best on Sinhala-script benchmarks have the most to lose, and
151 Romanized ability is nearly flat across a 42-fold parameter range, so selecting on formal-script scores
152 selects for the largest drop. The cheap quantity to log is absolute bits per byte, which tracks the
153 downstream gap ($\rho = -0.83$, $n = 10$), and not the bits-per-byte *gap*, which does not ($\rho = -0.15$,
154 Appendix G). The remedies are dull and available now: transliterate before inference, put Romanized
155 text in pretraining and instruction data, and evaluate in the register you serve.

156 6 Limitations

157 Everything here is one language, and the floor effect we describe may not transfer, although Khullar
158 et al. [2025] report compatible magnitudes for six South Asian languages. The Romanized side of the
159 downstream items is transliterated rather than human-typed, since no parallel resource exists at that
160 scale. Rather than assume that is neutral we measured it: our text is 0.17 to 0.24 bits per byte *harder*
161 than what native speakers typed for the same sentences, so our gaps are closer to upper bounds than
162 to lower bounds. We cannot exclude Sinhala-script evaluation material in pretraining, and everything
163 is zero-shot on open weights up to 14.7B, so this is what a user gets out of the box and not a ceiling.
164 Appendix I has the rest.

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A Related Work

Sinhala and low-resource evaluation. Sinhala is low-resource and its tooling has grown in a fragmented way [de Silva, 2019, Ranathunga and de Silva, 2022, Joshi et al., 2020]. Recent resources are almost entirely Sinhala-script: SinhalaMMLU [Pramodya et al., 2025], SOLD [Ranasinghe et al., 2025], the Sinhala split of Global PIQA [de Silva and Ranathunga, 2026, Chang et al., 2025] and encoder work such as Dhananjaya et al. [2022]. The only benchmark treating script as a variable is Rajapakse and Weerasinghe [2026], which we extend.

Romanization gaps elsewhere. Romanized South Asian text has been studied mainly as a resource and identification problem [Roark et al., 2020, Benton et al., 2025]. J et al. [2024] show romanization can *help* when models are adapted for it, the mirror image of our zero-shot setting. The closest same-language work trains a model for the register: Chamikara and Sumanathilaka [2026] fine-tune multilingual encoders for hate speech detection on Romanized Sinhala, and Hewapathirana and Sumanathilaka [2024] build a classifier over Romanized Sinhala social media. We instead ask what general-purpose generative models do out of the box. Closest in spirit is Khullar et al. [2025], who find weighted F1 falling by as much as 24 points on Romanized clinical queries. We add a paired item-level design with a competence screen, a tokenizer-independent account of the same models, and a test of whether the gap is a tokenization artifact.

Why not perplexity. Bits per byte exists so that models with different vocabularies can be compared [Gao et al., 2020], and Paloma adopts it for that reason [Magnusson et al., 2024]. Tokenizers treat languages very unequally [Petrov et al., 2023, Ahia et al., 2023] and over-segmentation costs accuracy on low-resource tasks [Nag et al., 2024]. To our knowledge ours is the first work to show that the substantive conclusions of a published benchmark change when the normaliser changes.

B Setup Details

Hardware and compute. The full study is roughly 80 hours of GPU wall clock. The intrinsic sweep is about 10 hours over 46,500 scored sequences, and the downstream run about 70 hours over 208,538 generations, of which roughly 25 hours are Phi-4-14B alone. Inference is fp16 on two T4 GPUs, except Phi-4-14B which ran on a single 16 GB RTX 4070 Ti SUPER. The transliteration control of §H adds 9,000 sequences in fp32 on 8 CPU cores in under an hour. No training was performed at any point.

Prompting. All prompts are zero-shot, greedily decoded with a 40-token limit, and answers are extracted by pattern match with anything unparseable scored as wrong. We chose one template from three candidates using a pilot of 3,240 generations over nine checkpoints, both script conditions and 20 items per dataset (Table 1). The direct template won on SinhalaMMLU, the largest set, and was the only candidate there that never produced unparseable output. An answer-first template scored 1.4 points higher on SOLD and 7.2 higher on Global PIQA, and we applied the direct template everywhere anyway, because a template chosen per dataset would confound task with prompt, and because 20 items cannot resolve differences of that size. What a paired comparison needs is not an optimal template but one that does not favour a script, and the evidence for that is the main run, where the unparseable rate never exceeds 0.03% in either condition for the competent cohort (Table 10). Twenty of the pilot’s SinhalaMMLU items, 0.3% of the benchmark, are also in the final evaluation set.

Statistics. Both conditions score the same items, so we use McNemar’s exact or continuity-corrected test [McNemar, 1947, Dror et al., 2018] and report discordant pair counts. Within each dataset we control the family-wise error rate over checkpoints with Holm’s procedure [Holm, 1979]. Intervals on gaps, macro-F1 differences, Matthews correlation differences and regression slopes come from 10,000 paired bootstrap resamples over items, except for the cross-checkpoint regressions of §4, which resample checkpoints. For SOLD we lead with Matthews correlation [Matthews, 1975, Chicco and Jurman, 2020] because the classes are imbalanced and accuracy sits close to the majority baseline. Item-level effects are also fitted as logistic regressions with checkpoint fixed effects and cluster-robust standard errors by item, which put the odds ratio for Romanized input on SinhalaMMLU at 0.70 ($p < 10^{-33}$) and give a significant difficulty interaction ($p < 0.001$).

Template	Accuracy %			Invalid %	
	S	R	both	S	R
<i>SinhalaMMLU</i>					
T1 direct (used)	28.3	25.0	26.7	0.0	0.0
T2 two-shot	18.3	28.9	23.6	10.6	2.8
T3 answer first	27.2	23.9	25.6	0.6	0.6
<i>SOLD</i>					
T1 direct (used)	51.1	51.1	51.1	10.6	8.9
T2 two-shot	42.8	44.4	43.6	10.6	5.6
T3 answer first	49.4	55.6	52.5	11.1	8.9
<i>Global PIQA</i>					
T1 direct (used)	48.9	40.6	44.7	4.4	3.3
T2 two-shot	46.1	48.3	47.2	13.9	0.6
T3 answer first	52.2	51.7	51.9	3.9	5.6

Table 1: The prompt-template pilot: 9 checkpoints, 3 templates, 2 script conditions, 20 items per dataset, 3,240 generations. Accuracy counts unparseable output as wrong. T1 was applied to all three datasets even though T3 scored higher on SOLD and Global PIQA, so that no dataset gets a different prompting regime. At 20 items per dataset none of these differences is resolvable, which is why the pilot is treated as a format check.

C Reproducing the Published Benchmark

Table 3 sets our recomputation beside every headline number of Rajapakse and Weerasinghe [2026] on its own 24 checkpoints. Per-checkpoint absolute deviations from the published perplexities have a median of 0.05% on Sinhala-script text and 0.004% on Romanized text, and 0.39% on mixed-script text once two outliers are excluded, Phi-4-14B and Mistral-Nemo-Base-2407. Everything in §3 is therefore the metric and not a different run. Table 4 applies Equation 1 per checkpoint, and Figure 3 shows the same split as stacked bars. Figure 2 shows the disagreement between the two metrics directly.

Two details are worth stating. Over all 31 checkpoints the normaliser share of the total falls to 39%, and for the six Qwen3.5 and Gemma checkpoints the normaliser term is slightly negative, because their tokenizers spend about the same number of bytes per token in both scripts. Those six also hold the five smallest ratios in the pool, with the four Qwen3.5 entries between 58-fold and 64-fold against a pool median of 294-fold, so the eye-catching part of a 312-fold figure is concentrated in the models whose tokenizers handle Sinhala worst. How completely the two metrics come apart also depends on the pool: on the 24 published checkpoints the rank correlation between perplexity and bits per byte is moderate ($\rho = 0.46$, $p = 0.02$), still leaving four fifths of bits-per-byte rank variance unexplained, and adding the seven newer checkpoints removes even that.

D Full Intrinsic Results

Table 5 gives all 31 checkpoints, sorted by Sinhala-script bits per byte. The two orderings disagree almost completely. Bits per word is the sharpest cross-script unit, because a sentence and its romanization contain the same number of words while Sinhala UTF-8 spends 2.65 bytes per character against 1.00 for Latin.

The mixed-script condition behaves quite differently from the Romanized one and reproduces the earlier finding that script handling tracks general ability. Sinhala-script and mixed-script bits per byte are strongly rank-correlated ($\rho = 0.94$) at a median gap of only 0.49 bits, against $\rho = 0.41$ for Romanized. Interleaving scripts is a mild perturbation. Replacing one entirely is not.

Table 6 gives the character n -gram reference points, fit on the same corpus under five-fold cross-validation so every score is out of sample, and Figure 4 plots them against the checkpoints. On Romanized Sinhala the best n -gram reaches 2.64 bits per byte and a character bigram 3.06, while the best checkpoint reaches 3.29 and the median 3.81, so no checkpoint beats the bigram. On the Sinhala script the picture is ordinary, with the character trigram at 1.16 and one checkpoint below it. The n -gram is a cheap in-domain model and the comparison favours it in that respect, so we read

Checkpoint	B	Hugging Face identifier	tokens / word	
			Sinh.	Rom.
OPT-350M●	0.35	facebook/opt-350m	14.1	2.4
Pythia-410M●	0.41	EleutherAI/pythia-410m	14.0	2.2
BLOOM-560M●	0.56	bigscience/bloom-560m	8.9	1.9
BLOOM-1B1●	1.10	bigscience/bloom-1b1	8.9	1.9
TinyLlama-1.1B-Chat‡●	1.10	TinyLlama/TinyLlama-1.1B-Chat-v1.0	9.8	2.4
Llama-3.2-1B●	1.20	meta-llama/Llama-3.2-1B	9.7	2.2
Cerebras-GPT-1.3B●	1.30	cerebras/Cerebras-GPT-1.3B	14.0	2.3
OPT-1.3B●	1.30	facebook/opt-1.3b	14.1	2.4
LaMini-GPT-1.5B‡●	1.50	MBZUAI/LaMini-GPT-1.5B	14.0	2.3
Qwen1.5-1.8B●	1.80	Qwen/Qwen1.5-1.8B	7.6	2.1
OPT-2.7B●	2.70	facebook/opt-2.7b	14.1	2.4
StableLM-Zephyr-3B‡●	2.80	stabilityai/stablelm-zephyr-3b	14.0	2.2
BLOOM-3B●	3.00	bigscience/bloom-3b	8.9	1.9
SmolLM3-3B‡●	3.10	HuggingFaceTB/SmolLM3-3B	9.5	2.1
Llama-3.2-3B●	3.20	meta-llama/Llama-3.2-3B	9.7	2.2
Qwen3.5-4B-Base	4.00	Qwen/Qwen3.5-4B-Base	4.3	2.0
Qwen3.5-4B‡	4.00	Qwen/Qwen3.5-4B	4.3	2.0
Mistral-7B-v0.3●	7.20	mistralai/Mistral-7B-v0.3	9.3	2.4
Zephyr-7B-beta‡●	7.20	HuggingFaceH4/zephyr-7b-beta	9.3	2.4
Qwen2-7B●	7.60	Qwen/Qwen2-7B	7.6	2.1
Qwen2-7B-Instruct‡	7.60	Qwen/Qwen2-7B-Instruct	7.6	2.1
Llama-3.1-8B●	8.00	meta-llama/Meta-Llama-3.1-8B	9.7	2.2
Llama-3.1-8B-Instruct‡	8.00	meta-llama/Llama-3.1-8B-Instruct	9.7	2.2
Hormoz-8B‡●	8.00	mann-e/Hormoz-8B	8.6	2.1
Minitron-8B-Base●	8.40	nvidia/Minitron-8B-Base	14.2	1.9
Gemma-7B●	8.50	google/gemma-7b	4.5	2.0
Qwen3.5-9B-Base	9.00	Qwen/Qwen3.5-9B-Base	4.3	2.0
Qwen3.5-9B‡	9.00	Qwen/Qwen3.5-9B	4.3	2.0
Gemma-2-9B	9.20	google/gemma-2-9b	4.5	2.0
Mistral-Nemo-Base-2407●	12.20	mistralai/Mistral-Nemo-Base-2407	14.2	2.2
Phi-4-14B‡●	14.70	microsoft/phi-4	9.5	2.1

Table 2: All evaluated checkpoints, with Hugging Face identifiers and tokenizer fertility on the parallel corpus. ‡ marks the eleven used downstream and ● the 24 that Rajapakse and Weerasinghe [2026] evaluated. Sinhala-script fertility varies by more than a factor of three across tokenizers, whereas Romanized fertility is nearly constant.

Quantity	Published	Ours
median PPL ratio, Sinh. to Rom.	312.3	312.3
its range	149.1 to 769.7	148.5 to 769.3
median PPL ratio, Sinh. to mixed	2.0	2.0
ρ (params, Sinh. PPL)	+0.09	-0.05
ρ (params, Rom. PPL)	-0.09	+0.11
ρ (Sinh. PPL, Rom. PPL)	+0.52	+0.53
ρ (Sinh. PPL, mixed PPL)	+0.96	+0.98
CV, Sinh. PPL	66.4%	67.8%
CV, Rom. PPL	62.3%	63.8%
median dev. , Sinh. PPL	—	0.047%
median dev. , Rom. PPL	—	0.004%

Table 3: Our recomputation against the published benchmark, on its own 24 checkpoints. Deviations are per-checkpoint absolute percentage differences from the published perplexities.

374 this as Romanized-condition checkpoints failing to exploit structure beyond what a character bigram
375 captures, rather than as a claim about what they know. Two BLOOM checkpoints sit near the bottom
376 of Table 5 on Sinhala-script text while being unremarkable on Romanized text, and BLOOM is the
377 only family in the pool with documented Sinhala pretraining exposure [BigScience Workshop et al.,
378 2023].

Checkpoint	ratio	bits	token-count normaliser		per-byte loss		norm. %
			bits	factor	bits	factor	
Qwen3.5-9B	58	5.86	-0.34	0.8	6.20	73.6	-6
Qwen3.5-9B-Base	58	5.87	-0.34	0.8	6.21	74.0	-6
Qwen3.5-4B-Base	63	5.98	-0.35	0.8	6.33	80.6	-6
Qwen3.5-4B	64	6.01	-0.36	0.8	6.36	82.4	-6
Gemma-7B●	149	7.21	-0.20	0.9	7.42	171.1	-3
Mistral-7B-v0.3●	175	7.45	2.72	6.6	4.73	26.6	37
Zephyr-7B-beta●	183	7.52	3.08	8.4	4.44	21.7	41
Qwen1.5-1.8B●	196	7.62	2.83	7.1	4.79	27.6	37
Qwen2-7B●	216	7.75	2.68	6.4	5.07	33.6	35
OPT-350M●	226	7.82	4.74	26.6	3.09	8.5	61
TinyLlama-1.1B-Chat●	232	7.86	2.95	7.7	4.90	29.9	38
LaMini-GPT-1.5B●	244	7.93	5.27	38.5	2.67	6.4	66
Qwen2-7B-Instruct	249	7.96	2.74	6.7	5.22	37.3	34
Phi-4-14B●	287	8.16	3.49	11.2	4.68	25.6	43
OPT-1.3B●	290	8.18	4.53	23.1	3.65	12.6	55
OPT-2.7B●	294	8.20	4.42	21.4	3.78	13.7	54
Llama-3.1-8B●	301	8.24	3.07	8.4	5.16	35.8	37
BLOOM-1B1●	323	8.34	4.72	26.3	3.62	12.3	57
Llama-3.1-8B-Instruct	325	8.35	3.28	9.7	5.07	33.5	39
Gemma-2-9B	357	8.48	-0.20	0.9	8.68	411.2	-2
Cerebras-GPT-1.3B●	364	8.51	4.39	21.0	4.11	17.3	52
BLOOM-560M●	389	8.60	4.93	30.6	3.67	12.7	57
Llama-3.2-3B●	417	8.70	3.33	10.1	5.37	41.4	38
Llama-3.2-1B●	419	8.71	3.42	10.7	5.29	39.0	39
StableLM-Zephyr-3B●	466	8.87	5.41	42.6	3.45	10.9	61
BLOOM-3B●	469	8.87	4.35	20.4	4.53	23.0	49
Pythia-410M●	482	8.91	4.45	21.9	4.46	22.0	50
Mistral-Nemo-Base-2407●	691	9.43	4.85	28.8	4.58	24.0	51
Hormoz-8B●	698	9.45	3.72	13.2	5.72	52.9	39
Minitron-8B-Base●	710	9.47	5.56	47.1	3.91	15.1	59
SmolLM3-3B●	769	9.59	4.50	22.6	5.09	34.0	47

Table 4: Equation 1 applied to every checkpoint, sorted by the reported perplexity ratio. “ratio” is Romanized over Sinhala-script perplexity and “bits” its base-2 logarithm, which the two terms partition exactly. The last column is the normaliser’s share of the total, and ● marks the 24 checkpoints of the published pool.

E Full Downstream Results

Table 7 gives all eleven checkpoints on SinhalaMMLU and SOLD, with macro-F1 alongside Matthews correlation. Table 8 breaks SinhalaMMLU into difficulty and domain strata pooled over the six competent checkpoints, Table 9 gives Global PIQA per checkpoint, and Table 10 gives the two controls of §4 per checkpoint, the unparseable rates and the ratio of Romanized to Sinhala-script input tokens. Figure 5 shows the SOLD result, where accuracy and Matthews correlation disagree about whether anything happened.

Two numbers in Table 7 are worth flagging. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat appear to improve under romanization, and both are artifacts of their failing to produce parseable output on the Sinhala-script side rather than script effects, which is what the competence screen exists to catch. The SOLD offensive-class rates behind the retreat toward triviality are 10.5% to 23.4% for Qwen2-7B-Instruct and 9.8% to 20.7% for Hormoz-8B.

F Every Refit of the Flattening Regressions

Table 11 lists every fit. One deserves explanation, because it is the natural way to state the result and it is a trap. Writing headroom as Sinhala-script accuracy minus chance, the paired loss fits headroom at $\text{loss} = 0.78 \times \text{headroom} - 1.06$ with $R^2 = 0.96$. But Sinhala-script accuracy appears on both axes, so a high fit is close to guaranteed. Shuffling the Romanized scores to destroy any association still yields a mean slope of 1.00 and a mean R^2 of 0.94, with 8.9% of permutations reaching $R^2 \geq 0.96$. We do not treat that fit as evidence, and we report it here because the same construction is easy to reach for in any robustness study. The levels form we do rely on returns a mean slope of 0.00 and a mean R^2 of 0.11 under the same null, and only 0.5% of permutations reach the observed fit. The cell-level refit is 0.48 when cells are weighted by item count. The two views are complementary

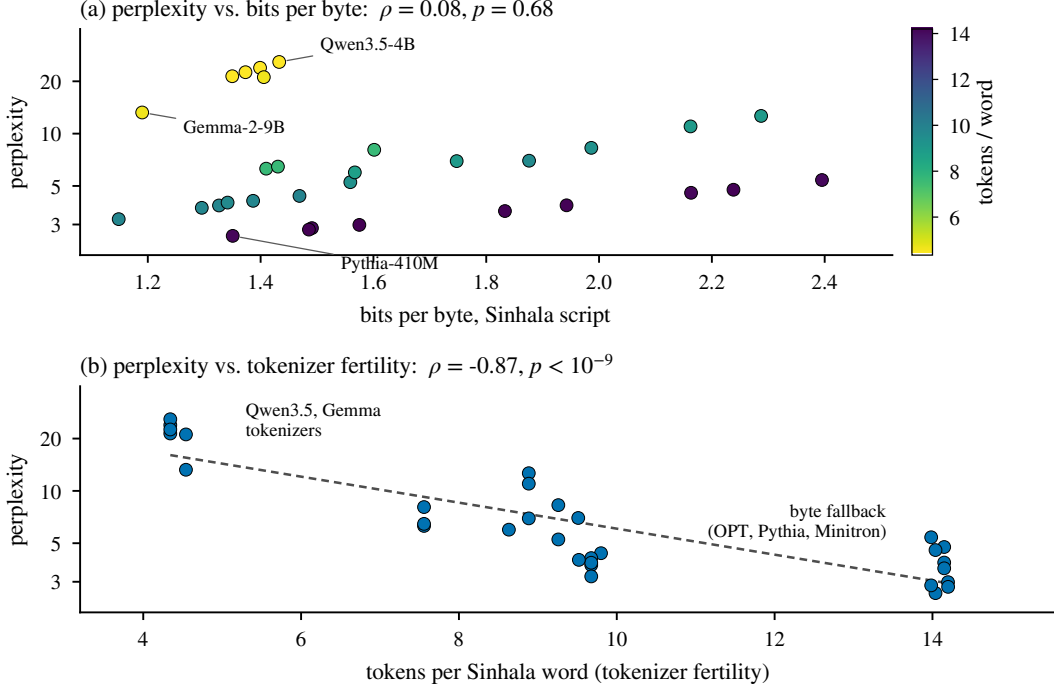


Figure 2: (a) Sinhala-script perplexity against Sinhala-script bits per byte over 31 checkpoints. The two are unrelated ($\rho = 0.08$, $p = 0.68$). (b) Perplexity against tokenizer fertility, which explains it almost fully ($\rho = -0.87$, $p < 10^{-9}$).

rather than independent, since they aggregate the same generations differently, but they agree on the shape and disagree usefully on the amount: content-level transfer is about twice model-level transfer.

G Do Intrinsic Metrics Track Downstream Fragility?

For the ten checkpoints measured both ways, absolute Sinhala-script bits per byte tracks the Sinhala-MMLU gap ($\rho = -0.83$, 95% CI -1.00 to -0.30 , leave-one-out -0.90 to -0.77) and the loss of Matthews correlation on SOLD ($\rho = -0.78$, CI -1.00 to -0.36 , leave-one-out -0.87 to -0.73), which is the flattening of §4 seen from the intrinsic side. The negative result is the more useful one. The bits-per-byte gap, the natural analogue of the degradation ratio prior work reports, shows no association ($\rho = -0.15$, CI -0.89 to $+0.63$), and neither does the perplexity ratio ($\rho = -0.52$). For the gap even the leave-one-out range straddles zero, so we cannot sign it.

All of this is exploratory: ten heterogeneous checkpoints, no held-out validation, no pre-registration, and intervals wide enough to reach -1.00 . The levels-versus-gaps contrast is stable under leave-one-out, but Table 12 is a hypothesis for a larger checkpoint set rather than a screening instrument.

H The Transliterator, and What It Costs

Our transliterator is a deterministic grapheme-level converter that parses Sinhala as an abugida and uses the digraph conventions speakers type rather than a scholarly scheme, so the dental stop becomes *th* and the labiodental approximant becomes *w*. It runs locally, covers every input, and collapses only the consonant distinctions casual Singlish also collapses.

Against human references. We validated it on 755,000 human-romanized items from the Swabhasha Resource Hub [Sumanathilaka et al., 2025]: 4,397 authentic social-media sentence pairs, 450,587 words each carrying several accepted spellings, and a 300,000-sentence machine-augmented cross-check. Scored against the closest accepted reference by character error rate, chrF [Popović, 2015] and exact match, ours has the lowest error rate of four methods on all three corpora, 0.182,

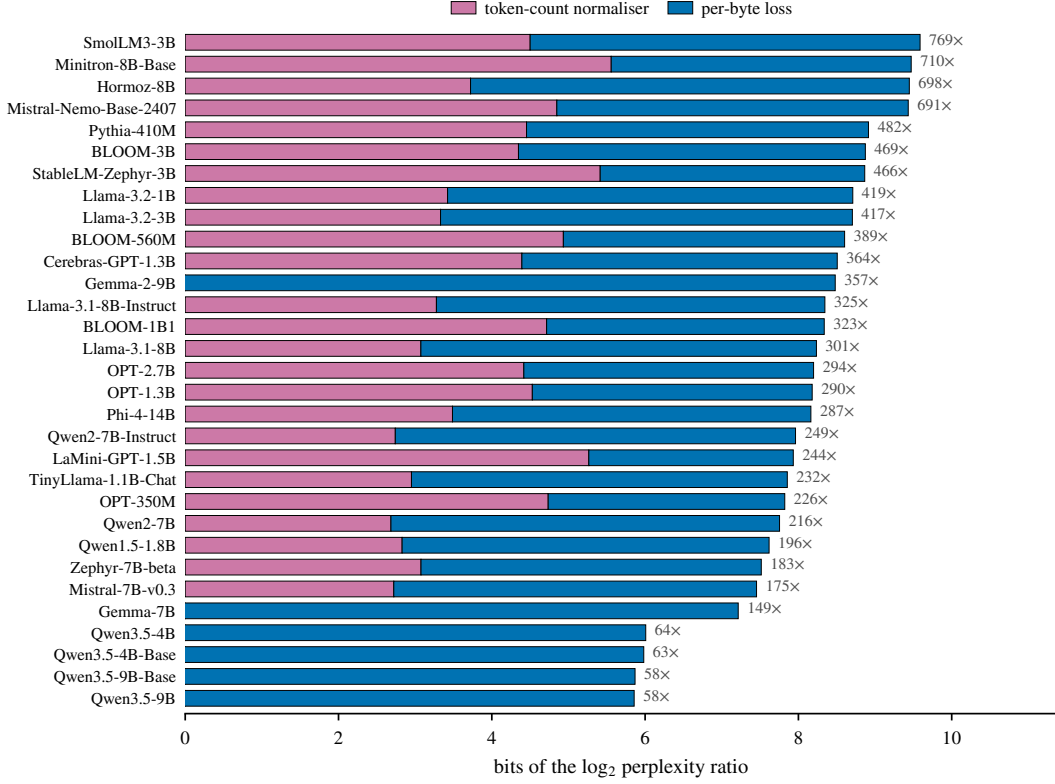


Figure 3: The split of Equation 1 per checkpoint. The two terms sum exactly to the reported gap in bits.

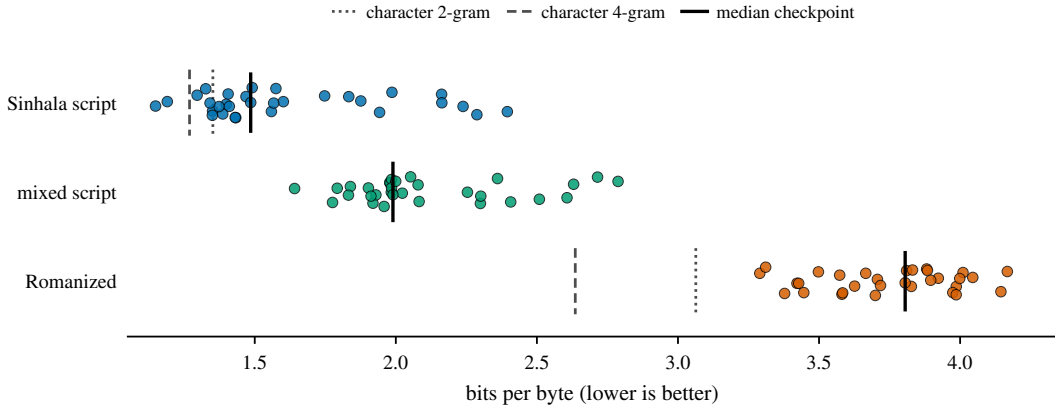


Figure 4: Checkpoint bits per byte against character n -gram references in both conditions. On Romanized Sinhala every checkpoint sits above the bigram.

0.121 and 0.112, and a paired Wilcoxon test separates it from Aksharamukha [Rajan, 2024], uroman [Hermjakob et al., 2018] and the web romanizer of de Silva and Ranathunga [2026] at $p < 10^{-90}$ on the two large corpora (Table 13). The residual difference from human typing is one convention rather than phonemic error. All four methods double long vowels far more often than people do, 0.81 per word against 0.12. Word by word against the human-typed side of the parallel corpus, 44.6% of our forms match exactly and a further 27.6% match once that convention is normalised, leaving 27.8% residual, and against the multi-reference lexicon of 449,598 words and 7,075,844 spellings, 88.6% of our spellings are ones a human actually supplied for that word, against 84.2% for the native speakers who typed the same sentences (Tables 14 and 15).

Checkpoint	Sinhala script				Romanized			Mixed	
	B	PPL	BPB	BPW	PPL	BPB	BPW	PPL	BPB
Llama-3.1-8B	8.0	3.22	1.148	16.3	971	3.584	22.1	6.36	1.641
Gemma-2-9B	9.2	13.23	1.190	16.9	4726	3.999	24.7	48.99	1.792
Llama-3.2-3B	3.2	3.74	1.296	18.4	1561	3.831	23.6	7.40	1.776
Llama-3.1-8B-Instruct‡	8.0	3.86	1.326	18.9	1256	3.718	22.9	7.94	1.839
Phi-4-14B‡	14.7	4.01	1.341	19.1	1149	3.446	21.3	8.08	1.832
Qwen3.5-9B-Base	9.0	21.39	1.350	19.2	1250	3.290	20.3	78.56	1.958
Pythia-410M	0.4	2.58	1.351	19.2	1245	3.707	22.9	4.91	1.978
Qwen3.5-9B‡	9.0	22.56	1.373	19.5	1308	3.311	20.4	84.24	1.990
Llama-3.2-1B	1.2	4.11	1.387	19.7	1719	3.881	24.0	8.54	1.903
Qwen3.5-4B-Base	4.0	23.92	1.399	19.9	1513	3.378	20.8	83.32	1.985
Gemma-7B	8.5	21.13	1.406	20.0	3139	3.805	23.5	66.00	1.930
Qwen2-7B	7.6	6.29	1.410	20.0	1357	3.498	21.6	14.11	1.919
Qwen2-7B-Instruct‡	7.6	6.46	1.431	20.3	1611	3.581	22.1	15.45	1.985
Qwen3.5-4B‡	4.0	25.84	1.433	20.4	1663	3.422	21.1	90.71	2.023
TinyLlama-1.1B-Chat‡	1.1	4.38	1.469	20.9	1014	3.923	24.2	8.92	1.980
Mistral-Nemo-Base-2407	12.2	2.80	1.486	21.1	1938	3.896	24.0	4.63	1.912
Cerebras-GPT-1.3B	1.3	2.86	1.491	21.2	1040	3.700	22.8	5.23	2.052
Mistral-7B-v0.3	7.2	5.26	1.559	22.2	921	3.884	24.0	10.15	1.984
Hormoz-8B‡	8.0	5.98	1.567	22.3	4176	4.045	25.0	13.58	2.083
Minitron-8B-Base	8.4	2.98	1.575	22.4	2119	3.428	21.2	5.08	2.000
Qwen1.5-1.8B	1.8	8.07	1.601	22.8	1585	3.573	22.1	17.60	2.079
BLOOM-3B	3.0	6.95	1.748	24.8	3258	3.626	22.4	15.90	2.254
OPT-2.7B	2.7	3.59	1.833	26.1	1053	3.974	24.5	6.26	2.300
SmolLM3-3B‡	3.1	6.98	1.876	26.7	5373	4.145	25.6	13.89	2.301
OPT-1.3B	1.3	3.87	1.942	27.6	1122	4.010	24.8	6.82	2.407
Zephyr-7B-beta‡	7.2	8.28	1.986	28.2	1516	4.167	25.7	15.76	2.360
BLOOM-1B1	1.1	11.00	2.162	30.7	3558	3.666	22.6	24.52	2.607
StableLM-Zephyr-3B‡	2.8	4.57	2.163	30.8	2129	3.986	24.6	7.51	2.509
OPT-350M	0.3	4.75	2.238	31.8	1076	3.986	24.6	8.15	2.630
BLOOM-560M	0.6	12.64	2.287	32.5	4915	3.810	23.5	28.01	2.715
LaMini-GPT-1.5B‡	1.5	5.41	2.395	34.1	1322	3.827	23.6	9.46	2.788

Table 5: All 31 checkpoints, sorted by Sinhala-script bits per byte. ‡ marks the eleven evaluated downstream. The perplexity and bits-per-byte orderings disagree almost completely: Gemma-2-9B is 2nd by bits per byte and 26th by perplexity, and Qwen3.5-9B-Base moves from 28th to 6th.

Model	Sinhala script		Romanized	
	BPB	# LMs <	BPB	# LMs <
character 1-gram	1.815	22	3.977	24
character 2-gram	1.352	7	3.063	0
character 3-gram	1.162	1	2.685	0
character 4-gram	1.269	2	2.636	0
character 5-gram	1.474	15	2.947	0
character 6-gram	1.680	21	3.381	3
best checkpoint	1.148	—	3.290	—
median checkpoint	1.486	—	3.805	—

Table 6: Held-out add-1 character n -gram models against the 31 checkpoints, five-fold cross-validated so every score is out of sample. “# LMs <” counts checkpoints achieving lower bits per byte than the n -gram model.

433 **What the substitution costs a model.** The Romanized side of the downstream items is ours rather
434 than human-typed, so we measured what that substitution costs. On the 500 parallel sentences, whose
435 Romanized side native speakers typed, our text is 0.174 to 0.240 bits per byte harder than theirs,
436 positive for all six checkpoints we could rerun, with a median of 0.219 (Table 16). Our condition is
437 therefore somewhat harder than authentic typing, which makes the downstream gaps we report closer
438 to upper bounds than to lower bounds. This cuts against what the spelling statistics alone would
439 suggest, and we report it because it bounds the claim rather than because it helps it. Two things limit
440 how far that control carries. It covers six of 31 checkpoints, all below 3.2B parameters, and none of
441 the six passes the downstream competence screen. For the checkpoints that do carry the downstream
442 result we instead test the same worry item by item on SOLD, and find that how closely an item’s
443 Romanized spellings match attested human spellings does not predict its Romanized penalty, with a
444 Spearman correlation of 0.02 ($p = 0.31$, $n = 2,447$) and terciles losing 5.0, 7.2 and 7.3 points from
445 least to most attested (Table 17). The intrinsic results do not share this limitation at all, since their

Checkpoint	SinhalaMMLU (6,879 items)						SOLD (2,500 items)					
	Acc. %					p_{Holm}	macro-F1		MCC		p_{Holm}	
	B	S	R	Δ	95% CI		S	R	S	R		
TinyLlama-1.1B-Chat	1.1	21.7	21.9	-0.2	[-0.5, 0.1]	.8273	6.4	24.2	-0.019	-0.043	1.0000	
LaMini-GPT-1.5B	1.5	0.9	18.8	-17.9	[-18.8, -16.9]	< .0001	5.1	28.3	0.000	0.024	1.0000	
StableLM-Zephyr-3B	2.8	22.9	24.0	-1.1	[-2.3, 0.1]	.4275	38.6	38.9	0.036	0.001	.9408	
SmolLM3-3B	3.1	25.1	25.5	-0.3	[-1.0, 0.4]	.8273	29.7	29.9	0.025	0.017	1.0000	
Qwen3.5-4B†	4.0	36.7	26.1	+10.6	[9.3, 12.0]	< .0001	60.2	55.1	0.219	0.127	.0018	
Zephyr-7B-beta	7.2	24.3	24.9	-0.6	[-1.7, 0.5]	.8273	39.7	39.6	0.060	0.060	1.0000	
Qwen2-7B-Instruct†	7.6	29.0	27.0	+2.0	[0.8, 3.2]	.0055	49.9	50.3	0.148	0.042	.0018	
Llama-3.1-8B-Instruct†	8.0	32.5	27.1	+5.4	[4.1, 6.7]	< .0001	60.1	52.6	0.259	0.105	< .0001	
Hormoz-8B†	8.0	26.0	25.2	+0.8	[-0.5, 2.0]	.8273	47.9	51.9	0.108	0.090	1.0000	
Qwen3.5-9B†	9.0	44.1	28.7	+15.5	[14.0, 16.9]	< .0001	63.1	52.0	0.280	0.069	< .0001	
Phi-4-14B†	14.7	28.7	27.2	+1.5	[0.2, 2.8]	.1286	54.4	50.8	0.152	0.076	.0084	

Table 7: All eleven downstream checkpoints, ordered by size. S is the Sinhala script, R is Romanized, Δ is the paired difference in points, p_{Holm} is McNemar’s test corrected across checkpoints within a task, and MCC is Matthews correlation with a paired-bootstrap p . † marks checkpoints passing the competence screen.

Stratum	Items	S %	R %	Δ	p_{Holm}
<i>by difficulty</i>					
Easy	1,851	37.0	29.2	7.9	< .0001
Medium	2,503	34.9	28.7	6.2	< .0001
Hard	2,525	27.7	23.4	4.3	< .0001
<i>by domain</i>					
Social Science	1,059	37.2	26.9	10.3	< .0001
Business Studies	463	34.4	27.1	7.3	< .0001
Other	1,014	31.7	24.6	7.0	< .0001
Stem	614	33.3	26.5	6.7	< .0001
Language	388	29.0	24.4	4.6	< .0001
Humanities	3,341	31.9	27.9	4.1	< .0001

Table 8: SinhalaMMLU by stratum, pooled over the six competent checkpoints. Item counts are per checkpoint and domains are ordered by loss.

446 Romanized side was typed by native speakers, and they show the same flattening. A human-written
447 Romanized version of even a few hundred SinhalaMMLU items would still settle the question better
448 than any of this.

449 I Full Limitations

450 **One language.** Everything here is Sinhala. The mechanism we describe, a floor effect that makes
451 robustness gaps proportional to formal-script ability, should generalise to any language with a
452 dominant Romanized register, but we have not shown that. Hindi, Nepali, Telugu, Urdu, Tamil and
453 Arabic dialects are the obvious tests, and Khullar et al. [2025] report compatible magnitudes for
454 Hindi, Nepali, Telugu, Kannada, Marathi and Punjabi.

455 **The Romanized side of the downstream sets is synthetic.** See §H. We transliterate rather than
456 collect human-typed Romanized versions, because no such parallel resource exists at this scale, and
457 we measured that our text is the harder end of the range rather than assuming it is neutral.

458 **Checkpoint size is confounded with everything else.** Our 31 checkpoints differ in family, to-
459 kenizer, training corpus and instruction-tuning status at the same time. The association between
460 parameter count and Sinhala-script bits per byte is not a scaling law, and our own fertility analysis is
461 direct evidence that family-level confounds are present.

462 **We cannot exclude contamination.** The training corpora of these checkpoints are not fully au-
463 ditable, so we cannot rule out that public evaluation sets, or Sinhala-script material closely related to
464 them, appeared during pretraining. If they did, the Sinhala-script side would be inflated and the gaps

Checkpoint	Acc. %		Δ	disc.	
	S	R		b/c	p_{Holm}
TinyLlama-1.1B-Chat	40	33	+7	12/5	1.0000
LaMini-GPT-1.5B	0	36	-36	0/36	<.0001
StableLM-Zephyr-3B	45	44	+1	15/14	1.0000
SmolLM3-3B	53	51	+2	14/12	1.0000
Qwen3.5-4B	57	49	+8	22/14	1.0000
Zephyr-7B-beta	48	50	-2	18/20	1.0000
Qwen2-7B-Instruct	45	42	+3	21/18	1.0000
Llama-3.1-8B-Instruct	53	48	+5	22/17	1.0000
Hormoz-8B	47	48	-1	13/14	1.0000
Qwen3.5-9B	53	45	+8	25/17	1.0000
Phi-4-14B	46	46	+0	18/18	1.0000

Table 9: Global PIQA, 100 items. “disc. b/c” are the discordant pair counts entering McNemar’s test. No checkpoint is above chance in the Sinhala-script condition and no comparison survives correction, so we treat this task as underpowered rather than as corroboration.

Checkpoint	MMLU inv. %		PIQA inv. %		τ
	S	R	S	R	
TinyLlama-1.1B-Chat	1.47	1.03	18	28	0.45
LaMini-GPT-1.5B	95.93	16.35	100	28	0.31
StableLM-Zephyr-3B	3.82	2.63	0	0	0.32
SmolLM3-3B	0.00	0.00	0	0	0.48
Qwen3.5-4B	0.00	0.00	0	0	0.72
Zephyr-7B-beta	2.11	0.64	0	0	0.49
Qwen2-7B-Instruct	0.03	0.01	0	0	0.51
Llama-3.1-8B-Instruct	0.00	0.00	0	0	0.45
Hormoz-8B	0.00	0.00	0	0	0.50
Qwen3.5-9B	0.00	0.00	0	0	0.72
Phi-4-14B	0.00	0.03	0	0	0.42

Table 10: Unparseable-output rates, and τ , the ratio of Romanized to Sinhala-script input tokens on SinhalaMMLU. τ is below one for every checkpoint, so the Romanized condition is the shorter input in every case.

we report overstated. This applies to any evaluation of pretrained models on public benchmarks and we have no way to bound it here.

The intrinsic-to-downstream link is exploratory. Ten checkpoints, no held-out validation, no pre-registration, and bootstrap intervals wide enough to reach -1.00 (§G).

The mixed-script condition is weaker evidence than the rest. That corpus is authentic mixed-script text rather than a transliteration of the parallel set, so it is not parallel to either script condition. It supports a descriptive statement about how mildly script mixing perturbs cost, and nothing paired.

Open weights only, and small ones. We evaluate open checkpoints up to 14.7B downstream. Frontier proprietary models are much stronger on Sinhala [Pramodya et al., 2025] and therefore have far more headroom, so the flattening slope predicts larger absolute losses for them. That prediction is untested here.

Zero-shot only. One template, zero-shot, greedy. This measures what a user gets out of the box, not the ceiling. Few-shot prompting, transliterate-then-infer, or Romanized fine-tuning could all narrow the gap, and we do not measure by how much.

Two checkpoints cannot do the task. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat produce unparseable output on most Sinhala-script items of at least one task. Their apparent improvement

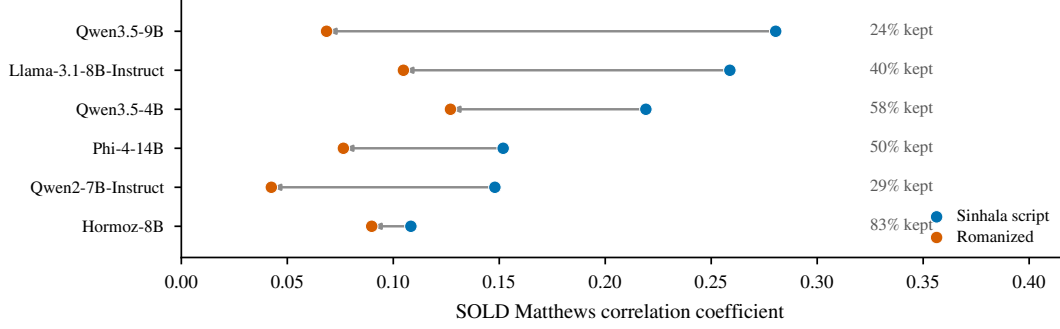


Figure 5: SOLD under romanization. Accuracy moves a few points against a 59.4% majority baseline while Matthews correlation loses a median 55% of its value.

Fit	Value
<i>across checkpoints, $n = 10$</i>	
levels: Romanized on native accuracy	
slope	0.221
95% boot. CI	[+0.113, +0.538]
leave-one-out	0.173 to 0.256
R^2	0.647
p , slope = 0	.0050
p , slope = 1	< .0001
<i>headroom form: loss on headroom</i>	
slope	0.779
95% boot. CI	[+0.457, +0.883]
leave-one-out	0.744 to 0.827
R^2	0.958
p , slope = 0	< .0001
p , slope = 1	.0050
<i>permutation null, Romanized scores shuffled</i>	
levels slope, mean	-0.000
levels R^2 , mean	0.113
levels $\Pr(R^2 \geq \text{observed})$	0.005
headroom slope, mean	+1.000
headroom R^2 , mean	0.936
headroom $\Pr(R^2 \geq \text{observed})$	0.089
<i>across benchmark cells, $n = 16$</i>	
levels: Romanized on native accuracy	
slope	0.451
95% boot. CI	[+0.359, +0.628]
leave-one-out	0.428 to 0.501
R^2	0.753
p , slope = 0	< .0001
p , slope = 1	< .0001
item-weighted slope	0.476

Table 11: Every refit of the two flattening regressions. The levels form regresses Romanized on Sinhala-script accuracy. The headroom form regresses the paired loss on Sinhala-script headroom above chance and is reported only to show why we do not rely on it.

481 under romanization is an artifact of that failure, not a script effect, which is why the competence
482 screen exists.

483 J Ethics

484 **Content warning.** §4 and Appendix E discuss an offensive-language dataset. We quote no offensive
485 text anywhere in this paper and report only aggregate metrics over it.

Intrinsic predictor	MMLU gap	SOLD Δ MCC
	ρ 95% CI	ρ 95% CI
Sinhala BPB	-0.83 [-1.00, -0.30]	-0.78 [-1.00, -0.36]
Romanized BPB	-0.85 [-1.00, -0.43]	-0.88 [-1.00, -0.40]
Sinhala BPW	-0.83 [-1.00, -0.31]	-0.78 [-1.00, -0.36]
BPB gap	-0.15 [-0.89, +0.63]	-0.42 [-0.83, +0.33]
PPL ratio	-0.52 [-0.94, +0.28]	-0.41 [-0.96, +0.40]
parameters	+0.58 [-0.06, +0.91]	+0.47 [-0.14, +0.90]

Table 12: Spearman correlations between intrinsic measurements and downstream script effects, over the ten parseable checkpoints, with bootstrap intervals. Levels track the effect, gaps and ratios do not. The intervals are wide because $n = 10$, which is why this is exploratory.

Method	CER $\downarrow$			chrF	Exact	Cov.
	sent.	word	aug.	sent.	word	%
Ours	0.182	0.121	0.112	67.8	33.2	100.00
Web tool $v \rightarrow w$	0.182	0.133	0.114	67.8	30.9	99.67
Aksharamukha	0.191	0.148	0.139	63.7	23.9	100.00
Web tool, raw	0.197	0.176	0.153	63.4	20.6	99.67
uroman	0.228	0.228	0.210	55.7	9.7	100.00

Table 13: Transliteration quality against human references: 4,397 authentic social-media sentence pairs, 450,587 multi-reference words, and a fixed 300,000-sentence sample of a machine-augmented set. Coverage is the share of inputs the method produced any output for.

Data. All datasets are public research releases used for evaluation only. Global PIQA forbids training on it or on synthetic data seeded from it, and we do neither. SinhalaMMLU is released under a licence that forbids distributing adaptations, and its authors asked that the full set not be made public, so our Romanized version of it stays local. SOLD contains offensive social media text, already pseudonymised upstream with user mentions replaced by placeholders, and we neither collected nor re-annotated any of it. We add no new human annotation and no new data collection, so no ethics approval was required.

Models. All 31 checkpoints were used for evaluation only, with no fine-tuning, redistribution or derivative model release, which is within the terms of each family’s licence, including the Llama Community License, the Gemma Terms of Use and the BigScience RAIL licence.

Risk. The practical risk our results speak to is deployment. A moderation or support system validated on Sinhala-script text and serving Romanized users will look far better in evaluation than in the field, and on SOLD that failure is invisible in accuracy while more than half the usable signal is gone. Publishing the size of that gap, and a screen that stops averaging from hiding it, is the useful response. Our transliterator collapses several Sinhala consonant distinctions, matching casual Singlish practice, so it is a tool for building evaluation data and not for rendering names or other text where those distinctions carry meaning.

K Licences and Released Artifacts

What we release. The anonymous repository at <https://anonymous.4open.science/r/XXXXXXXXX> contains the transliterator, the evaluation harness, all analysis code, the frozen result files and the scripts that regenerate every number, table and figure in this paper. The transliterator is also on PyPI, but naming the package would identify us, so the name is withheld until the camera-ready version.

Datasets. SinhalaMMLU [Pramodya et al., 2025] is CC BY-NC-ND 4.0 and gated. We received the full set from its authors, who ask that it not be redistributed, and the no-derivatives clause forbids distributing our Romanized adaptation, so we release the transformation code and the item identifiers instead of the items. SOLD [Ranasinghe et al., 2025] carries no explicit licence file and we use

Text	tokens	cov. %	attested %	
			raw	norm.
parallel, typed by people	2,797	96.1	83.4	84.2
parallel, ours	2,905	96.0	64.3	88.6
SOLD, ours	38,544	90.3	59.5	87.0
Global PIQA, ours	2,244	96.3	63.9	86.9

Table 14: Share of word tokens whose Romanized spelling is one a human supplied for that Sinhala word in the Swa-bhasha lexicon. “tokens” counts the word tokens the lexicon covers and “cov.” the share of all Sinhala word tokens that is. “norm.” collapses runs of the same Latin vowel on both sides. The first row is the reference point: it applies the same measure to text native speakers typed.

Ours	Human	n
<i>mee</i>	<i>meh</i>	33
<i>man</i>	<i>mn</i>	30
<i>daen</i>	<i>dan</i>	28
<i>naae</i>	<i>na</i>	26
<i>mata</i>	<i>matta</i>	19
<i>ee</i>	<i>a</i>	15
<i>naae</i>	<i>nah</i>	12
<i>ban</i>	<i>bn</i>	12
<i>nam</i>	<i>nm</i>	11
<i>gaena</i>	<i>gana</i>	10
<i>thamayi</i>	<i>thamai</i>	10
<i>waedak</i>	<i>wadak</i>	9

Table 15: The twelve most frequent word-level disagreements with the human-typed side of the parallel corpus that survive normalising the long-vowel convention. Most are vowel reduction or abbreviation on the human side.

its published test split for evaluation only. The Sinhala split of Global PIQA [Chang et al., 2025, de Silva and Ranathunga, 2026] is CC BY-SA 4.0 and forbids training, and we use it for evaluation only. The parallel and mixed-script corpora come from Rajapakse and Weerasinghe [2026], and the human-romanized reference data from the Swa-bhasha Resource Hub [Sumanathilaka et al., 2025].

Tools. Aksharamukha [Rajan, 2024] v2.3 is AGPL-3.0. As its licence requires, we state that this project uses the universal romanizer software ‘uroman’ written by Ulf Hermjakob, USC Information Sciences Institute (2015-2020) [Hermjakob et al., 2018].

NeurIPS Paper Checklist

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: The abstract and Section 1 state the three claims the paper supports, namely that half the published degradation ratio is the token-count normaliser (Section 3), that romanization compresses rather than uniformly lowers downstream ability (Section 4), and that the metric a benchmark reports decides which of these is visible (Section 5). Scope is one language and eleven open instruction-tuned checkpoints, stated in Section 2 and bounded in Section 6.

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
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Checkpoint	bits per byte				PPL	
	B	Sinh.	human	ours	Δ	ratio
OPT-350M	0.35	2.238	3.986	4.195	+0.208	0.85
Pythia-410M	0.41	1.350	3.707	3.937	+0.230	0.80
BLOOM-560M	0.56	2.287	3.810	4.029	+0.220	0.62
TinyLlama-1.1B-Chat‡	1.10	1.468	3.922	4.096	+0.174	0.73
Qwen1.5-1.8B	1.80	1.601	3.573	3.791	+0.218	0.73
SmolLM3-3B‡	3.10	1.878	4.147	4.387	+0.240	0.75
median	—	—	—	—	+0.219	0.74

Table 16: Our transliteration against human typing on the same 500 sentences. Δ is ours minus human bits per byte, positive meaning our text is harder. “PPL ratio” is our perplexity over the human perplexity, below one for every checkpoint even though bits per byte is higher, because our spellings occupy more tokens. The Sinhala-script and human columns reproduce the recorded run to within 0.11%. ‡ marks checkpoints also evaluated downstream. Six of 31 checkpoints, all below 3.2B and none passing the competence screen, so this bounds the direction rather than settling it.

Attest.	items	att. %	S %	R %	Δ	95% CI
low	822	40.8	61.0	56.0	5.0	[3.4, 6.6]
middle	925	60.7	61.9	54.7	7.2	[5.6, 8.8]
high	700	78.9	60.4	53.2	7.3	[5.4, 9.1]

Table 17: SOLD items split into terciles by how often their Romanized spellings are attested in the human lexicon, pooled over the five competent checkpoints. If our transliterator’s idiosyncrasies drove the Romanized penalty, the low-attestation tercile would lose most. It loses least. The tercile with the smallest gap also has the lowest Sinhala-script accuracy, which the flattening of §4 already predicts.

- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

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Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 6 lists the binding limitations, and Appendix I gives the full set, including the single language, the synthetic Romanized side of the downstream items and what we measured that substitution to cost, the confounding of scale with family and tokenizer, unauditable pretraining corpora, and the exploratory status of the intrinsic-to-downstream link.

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Justification: The paper proves no theorems. The one analytic step, the exact split in Equation 1, is an algebraic rearrangement of the definition of perplexity, it is derived in full in Section 3, and we verify it numerically on our own measurements to 5×10^{-15} bits.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 2 and Appendix B give the checkpoint list with Hugging Face identifiers and revisions, the corpora and splits, the prompt templates, the decoding settings, the answer-extraction rule, the competence screen, and every statistical procedure. Appendix C additionally reproduces the published perplexities we build on, to a median of 0.05%.

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 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All analysis code, the transliterator, the frozen result files and the scripts that regenerate every number, table and figure in this paper are in the anonymous repository at <https://anonymous.4open.science/r/XXXXXXX>. One dataset cannot be redistributed: SinhalaMMLU is licensed CC BY-NC-ND 4.0 and its authors asked that the full set stay private, so we release the transformation code and the item identifiers instead of the items (Appendix K).

Guidelines:

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6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: There is no training in this work, so the relevant details are evaluation details: zero-shot prompting, greedy decoding with a 40-token limit, one template chosen on a

documented pilot, no item sampling, and fp16 inference. All of these are in Section 2 and Appendix B.

Guidelines:

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7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Every paired comparison is tested with McNemar’s test under Holm correction within a task and reported with its discordant pair counts, and every interval is a 95% confidence interval from 10,000 paired bootstrap resamples, with the resampling unit stated in each case (Section 2). Regression slopes carry bootstrap intervals, leave-one-out ranges and permutation tests, and Appendix F reports one fit we discard as an algebraic artifact after its own permutation null reproduced it.

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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 2 reports 80 GPU hours of wall clock in total, about 10 for the intrinsic sweep over 31 checkpoints and about 70 for the 208,538 downstream generations, of which roughly 25 are one 14.7B checkpoint. Appendix B names the hardware, two T4 GPUs in fp16 with a single 16 GB RTX 4070 Ti SUPER for the largest checkpoint, and the CPU cost of the transliteration control.

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Answer: [Yes]

Justification: Section 5 and Appendix J discuss both directions: the positive impact is that a safety system validated on the formal script and serving Romanized users can now be shown to be far worse in the field than in evaluation, and the negative one is that our transliterator deliberately collapses consonant distinctions, so it suits building evaluation data and not rendering names.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We release no model weights, no scraped data and no generative artifact. The released code is a deterministic rule-based transliterator and the analysis pipeline, and the offensive-language corpus we measure on is an existing pseudonymised research release that we do not redistribute.

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Answer: [Yes]

Justification: The new assets are the Sinhala-to-Latin transliterator, the Romanized evaluation conditions and the analysis pipeline, all documented in Appendix H and in the README of the anonymous repository at <https://anonymous.4open.science/r/XXXXXXX>, which states the licence, the intended use and the known limitations of each.

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Answer: [N/A]

Justification: We ran no crowdsourcing and no study with human subjects. The human-typed Romanized text we validate against comes from existing public research releases collected by their own authors.

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Answer: [N/A]

Justification: Language models are the object of study here rather than a component of the method. The method itself is a deterministic transliterator, a tokenizer-independent metric and a paired statistical design, none of which involves an LLM.

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